Island Chill MES

ERPNext Integration & Administration Guide

Complete reference for connecting the Manufacturing Execution System to ERPNext — covering Stock, Inventory, Users, Warehouse, BOM, Work Orders, and all operational modules.

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Part A — Integration Architecture

A.1 How the MES Connects to ERPNext

Island Chill MES integrates with ERPNext through the REST API v2 layer. The MES acts as an API client: it reads master data (items, warehouses, users, BOMs) from ERPNext and writes transactional data (stock movements, quality inspections, work order progress) back to ERPNext in real time or on a configurable sync schedule.

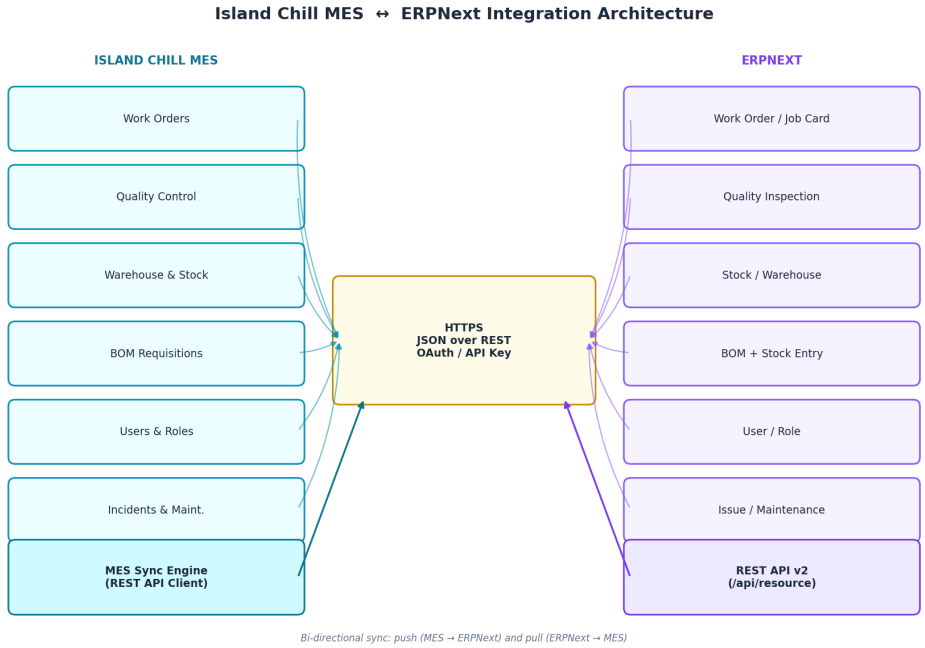


Figure A-1: MES ↔ ERPNext Integration Architecture

Integration principles:

• ERPNext is the system of record for master data: items, warehouses, UOMs, BOMs, and user accounts. The MES mirrors this data locally for fast, offline-tolerant operation.

• The MES is the system of engagement for shop-floor execution: operators perform work in the MES; results are pushed to ERPNext as Stock Entries, Quality Inspections, and Work Order updates.

• Bi-directional sync: master data flows ERPNext → MES (pull); execution data flows MES → ERPNext (push).

• Idempotent writes: every push carries a unique external reference so retries never create duplicates.

• Queue-based delivery: if ERPNext is unreachable, transactions are queued locally and delivered when connectivity returns.

A.2 Module-to-DocType Mapping

Every MES module maps to one or more ERPNext DocTypes. The table below is the authoritative reference used by the sync engine; Figure A-2 shows the same mapping visually.

MES Module	ERPNext DocType(s)	Sync Direction	Frequency
Users & Roles	User, Role, Role Profile	ERPNext → MES	On change / hourly
Warehouse Zones	Warehouse, Warehouse Type	ERPNext → MES	On change / hourly

Stock / Inventory	Item, Stock Ledger Entry, Bin	Bi-directional	Real-time push; hourly pull
Products & Variants	Item, Item Group, UOM	ERPNext → MES	On change / hourly
BOMs	BOM, BOM Item, BOM Operation	ERPNext → MES	On change
Work Orders	Work Order, Job Card	Bi-directional	Real-time
BOM Requisitions	Stock Entry (Material Transfer), Material Request	MES → ERPNext	Real-time
Quality Control	Quality Inspection, Quality Inspection Template, QI Reading	MES → ERPNext	Real-time
Incidents	Issue, Warranty Claim (optional)	MES → ERPNext	Real-time
Maintenance	Maintenance Schedule, Maintenance Visit, Asset Maintenance	Bi-directional	Daily / real-time
Filling Lines / Equipment	Workstation, Asset, Asset Movement	ERPNext → MES	On change

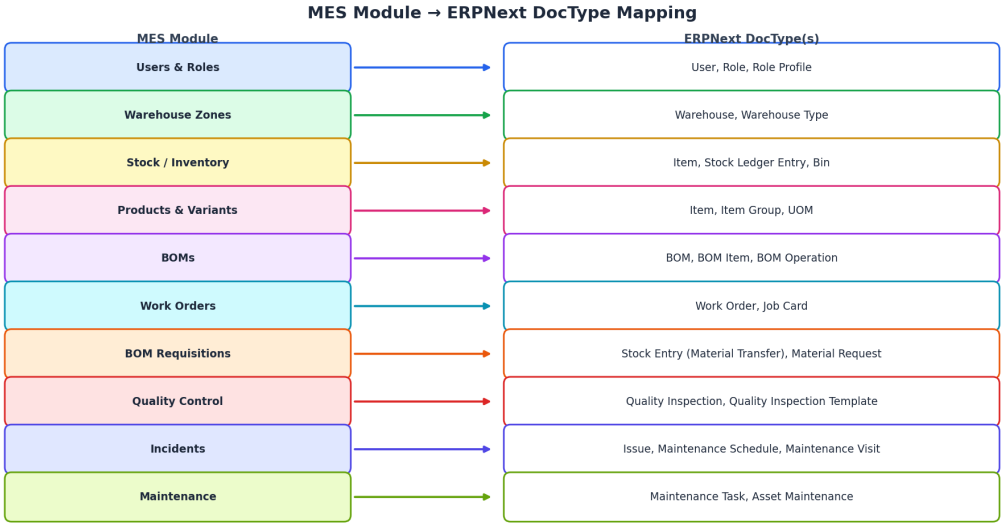


Figure A-2: MES Module → ERPNext DocType Mapping

Part B — ERPNext Setup for Administrators

This part is written for the Administrator who configures ERPNext. Each section gives the exact navigation path in ERPNext, the fields to set, and how the data appears inside the MES. Complete these sections in order — later sections depend on earlier ones.

B.1 Prerequisites & Access

Before configuring, confirm the following:

Requirement	Details
ERPNext version	v14 or v15 (Frappe Framework v14/v15). Cloud (frappecloud.com) or self-hosted both work.
Admin account	A user with the System Manager role in ERPNext.
Enabled modules	Stock, Manufacturing, Quality, Maintenance, HR (for users). Enable via Settings → Enable Modules.
HTTPS	ERPNext must be reachable over HTTPS for the MES to connect (required for API key auth).
MES admin access	Log into the MES as a user with the Administrator or Factory Manager role.

B.2 Company & Global Defaults

The company record is the root of all ERPNext data. If a company already exists you can reuse it.

Navigation: Settings → Company → New (or open existing)

Field	Recommended value	Notes
Company Name	Island Chill Beverages Ltd	Shown on all documents and reports.
Abbreviation	ICB	Used as prefix on warehouse names and naming series.
Default Currency	Your local currency (e.g. USD / EUR / SCR)	Must match MES currency setting.
Chart of Accounts	Standard template for your country	Needed before any stock transactions.
Default Warehouse (Stock Settings)	ICB-Stores — Raw Material	Set under Stock Settings → Defaults.
Naming Series	BUY-.##### / SAL-.#####	Use ERPNext defaults unless your finance team dictates otherwise.

MES link: the MES reads company, currency, and default warehouse from ERPNext and displays them in Admin → Settings. No MES-side configuration is needed for this section.

B.3 Warehouse Setup (Mapped to MES Zones)

Warehouses in ERPNext correspond to storage locations. The MES Warehouse page groups them into zones (Refrigerated, Ambient, Frozen, Quarantine). Create one ERPNext warehouse per physical store location, then the MES mirrors them as zones.

Navigation: Stock → Warehouse → New (create each warehouse below)

ERPNext Warehouse Name	Parent	Type	MES Zone Mapping
ICB-Stores — Raw Material	ICB-Stores	Ambient	Ambient → Raw Materials
ICB-Stores — Packaging	ICB-Stores	Ambient	Ambient → Packaging
ICB-Stores — Refrigerated	ICB-Stores	Refrigerated	Refrigerated
ICB-Stores — Frozen	ICB-Stores	Frozen	Frozen
ICB-Production — WIP	ICB-Production	Ambient	Ambient → Production staging
ICB-Quarantine	Root	Quarantine	Quarantine (blocked stock)
ICB-Finished Goods	Root	Ambient	Finished Goods dispatch
ICB-Scrap / Rework	Root	Ambient	Non-conforming items

How to create: Stock → Warehouse → New. Set Warehouse Name, choose Parent Warehouse, and tick Is Group only for group nodes (ICB-Stores, ICB-Production). Leave Account mapped to the default stock-in-hand account.

MES link: Warehouse Type in ERPNext (Refrigerated / Ambient / Frozen / Quarantine) maps to the MES zone temperature profile. Set Warehouse Type via Stock → Warehouse Type. The MES reads live stock from the Bin doctype per warehouse.

B.4 Units of Measure (UOM)

Every quantity in the MES (BOM lines, requisitions, stock) uses UOMs defined in ERPNext. Create the full set before creating items.

Navigation: Stock → UOM → New

UOM	Must be whole number?	Used for
Bottle	Yes	Finished goods count
ml / L	No	Liquids (water, juice base)
g / kg	No	Additives, mineral blends
pc	Yes	Caps, labels, bottles as components
Pack	Yes	Shrink-wrapped multipacks
CO2 kg	No	Carbonation gas

Conversions: set UOM conversion factors under Stock → UOM Conversion Factor (e.g. 1 L = 1000 ml, 1 Pack = 24 Bottles). The MES uses these factors when scaling BOM quantities to batch size.

B.5 Items: Raw Materials, Packaging, Finished Goods

Items are the foundation of stock and BOMs. Create items in three batches: raw materials and additives, packaging, then finished goods. Use the MES item codes shown in the app (e.g. RM-Water, PKG-330PET) so the sync engine matches records without ambiguity.

Navigation: Stock → Item → New

Item	Item Code	Item Group	Stock UOM	Maintain Stock?	MES Use
Purified Water	RM-Water	Raw Material	ml	Yes	BOM input
Mineral Blend	ADD-Mineral	Raw Material	g	Yes	BOM input
CO2 Gas	RM-CO2	Raw Material	CO2 kg	Yes	Carbonation
Sugar Syrup	RM-Syrup	Raw Material	L	Yes	Flavored variants
330ml PET Bottle	PKG-330PET	Packaging	pc	Yes	BOM input
500ml PET Bottle	PKG-500PET	Packaging	pc	Yes	BOM input
28mm Screw Cap	PKG-Cap28	Packaging	pc	Yes	BOM input
Island Water Label	PKG-LabelIW	Packaging	pc	Yes	BOM input
Island Water 330ml	FG-IW-330	Finished Goods	Bottle	Yes	Saleable product
Island Chill Original 500ml	FG-IC-500	Finished Goods	Bottle	Yes	Saleable product

Key fields on the Item form:

Field	Set to
Item Code	Exactly the MES code (e.g. RM-Water). This is the join key for sync.
Item Name	Human-readable name shown in the MES pick lists.
Item Group	Raw Material / Packaging / Finished Goods / Additive (see B.6).
Stock UOM	The UOM from B.4.
Maintain Stock	Ticked for everything physical. Unticked only for services.
Include Item In Manufacturing	Ticked for raw materials and packaging (makes them selectable on BOMs).
Default Warehouse	The store warehouse from B.3 (e.g. ICB-Stores — Raw Material).
Valuation Method	FIFO (standard for beverages).
Shelf Life / Expiry	Set 'Has Batch No' + 'Automatically Create Batch' and enter shelf life in days — drives MES expiry alerts.
Reorder Level	Enter reorder point — the MES low-stock alert mirrors ERPNext's reorder notification.

B.6 Item Groups & Categories

Item groups organize the catalog. The MES product families (Island Water, Island Chill, etc.) map to ERPNext item groups, and MES variants map to items.

Navigation: Stock → Item Group → New. Create this tree:

All Item Groups
- ├─ Raw Material (Purified Water, Mineral Blend, CO2, Syrup)
- ├─ Packaging (Bottles, Caps, Labels, Shrink Wrap)
- └─ Finished Goods
  - ├─ Island Water (Still Water family)
  - ├─ Island Chill (Flavored range)
  - └─ Island Sparkle (Carbonated range)

MES link: MES Admin → Products reads the Finished Goods subtree; each leaf item becomes a product variant with its SKU, UOM and default warehouse.

B.7 Users & Roles in ERPNext

ERPNext users authenticate against the MES via shared accounts. The recommended pattern is one ERPNext user per MES user, with ERPNext roles matching the MES role's data-access needs. The MES keeps its own fine-grained permission matrix (22 submodules); ERPNext roles only control what the API can read/write.

Navigation: Settings → User → New; roles under Settings → Role

MES Role	ERPNext Role(s) to assign	Purpose
Administrator	System Manager	Full ERPNext configuration and API access.
Factory Manager	Manufacturing Manager, Stock Manager	Work orders, stock entries, reports.
Planner	Manufacturing User, Stock User	Create Work Orders and BOMs, view stock.
Production Supervisor	Manufacturing User	Job cards and production progress.
Operator	Manufacturing User (restricted to Job Card)	Execute job cards only.
QC Inspector	Quality Manager	Create and submit Quality Inspections.
Maintenance Supervisor	Maintenance Manager, Maintenance User	Maintenance schedules, visits, issues.
Warehouse Supervisor	Stock Manager, Stock User	Stock entries, deliveries, adjustments.
Ingredient Mixer	Stock User	Material transfer stock entries (store → production).

Creating each user: Settings → User → New. Set email (used as the login id), first/last name, and assign roles. Under the API Access section, tick Generate Keys to create the user's API key/secret (see Part C). Disable 'Send Welcome Email' for kiosk/shared accounts.

Sync rule: The MES user list (Admin → Users) syncs from ERPNext users that have the 'In MES' custom field ticked. Add a custom field on User (Settings → Customize Form → User) named in_mes (Check type) to control which ERPNext users appear in the MES.

B.8 Bills of Materials (BOM)

The BOM in ERPNext is the authoritative recipe. The MES BOM pages and BOM Requisitions read from it. Create one BOM per finished-good variant, at the base quantity you produce per batch.

Navigation: Manufacturing → BOM → New

BOM Field	Example value	Notes
Item	FG-IW-330 (Island Water 330ml)	The finished good this BOM produces.
Quantity	1000	Base batch quantity. MES scales lines to the WO quantity.
UOM	Bottle	Match the item's stock UOM.
Company	Island Chill Beverages Ltd	From B.2.

Currency / Rate	Leave default	Optional costing; MES uses it for cost reports.

BOM Explosion Items (the raw material / packaging lines):

Item	Qty	UOM
RM-Water — Purified Water	330	ml
ADD-Mineral — Mineral Blend	0.05	g
PKG-330PET — 330ml PET Bottle	1	pc
PKG-Cap28 — 28mm Screw Cap	1	pc
PKG-LabelIW — Island Water Label	1	pc

Set Scrap % on each line to match the MES scrap allowance. Set an Operation if you use routing; otherwise submit the BOM directly. The MES syncs the submitted BOM automatically — draft/unsaved BOMs do not sync.

B.9 Stock Entry Types & Material Flow

The MES BOM Requisition workflow is powered by Stock Entries. When the Ingredient Mixer delivers ingredients to production, the MES posts a Material Transfer stock entry in ERPNext. No custom setup is needed — but naming series should be confirmed:

Navigation: Stock → Stock Entry Type (view the standard types)

Stock Entry Type	Used by MES when
Material Receipt	Goods receipt to warehouse (Warehouse → Goods Receipt).
Material Transfer	BOM Requisition delivered (store → production WIP).
Manufacture	WO completed — consumes BOM items, produces finished goods.
Material Issue	Scrap / write-off from QC rejection or incident.
Repack	Repacking into multipacks, if used.

Material flow in the MES: Requisition picking reserves stock (a Draft Stock Entry is created); when status becomes delivered the entry is submitted and stock moves from the store warehouse to ICB-Production — WIP. If the requisition is rejected, the draft is cancelled and reservation released.

B.10 Work Stations & Operations (Filling Lines)

Filling lines in the MES map to ERPNext Workstations. This enables capacity planning and job card timings in ERPNext.

Navigation: Manufacturing → Workstation → New

Workstation	MES Line	Electricity Cost/hr	Rent Cost/hr
Line 1 — Filler 500ml	Line 1	0.00 (or actual)	0.00
Line 2 — Filler 330ml	Line 2	0.00	0.00
Line 3 — Canning	Line 3	0.00	0.00
Line 4 — Mixer / Blender	Line 4	0.00	0.00

Operations: Manufacturing → Operation → New. Create 'Bottle Filling', 'Capping & Labelling', 'Packaging'. Link each operation to a workstation and default time (e.g. 60 min per 1000 bottles). The MES reads workstations and shows line status on the Flow Map and Digital Twin pages.

B.11 Quality Inspection Setup

MES QC checks (parameter, range, reading, pass/fail) map to ERPNext Quality Inspections. Define templates per product so inspectors in the MES can submit results without re-entering parameters.

Step 1 — Quality Inspection Template Navigation: Quality → Quality Inspection Template → New

Template Field	Example
Template Name	Island Water — Filled Bottle Check
Item	FG-IW-330
Reading Fields	Brix (°) range 0–0.2 · pH range 6.8–7.4 · Fill Volume (ml) 325–335 · Torque (N·m) 12–18 · CO2 Volume 3.5–4.5
Formula / Criteria	Acceptance based on min–max per field

Step 2 — Enable inspection on items. Open each finished-good item (Stock → Item) and under Quality Inspection set 'Inspection Required Before Delivery' = Yes and link the template.

Step 3 — Non-conformance. For failed checks, the MES creates a Stock Entry of type Material Issue to ICB-Scrap / Rework and links it to the inspection record.

B.12 Maintenance & Issue Setup

MES Incidents become ERPNext Issues (for tracking and SLA); MES Maintenance Schedules become ERPNext Maintenance Schedules + Visits.

Navigation: Maintenance → Maintenance Schedule → New; Support → Issue

MES Record	ERPNext Record	Setup needed
Incident (equipment)	Issue + linked Asset	Register equipment as Asset (Assets → Asset) with serial number matching the MES equipment ID.
Incident (quality/safety)	Issue (type = Quality / Safety)	Create Issue Types under Support → Issue Type.
Maintenance Task (pre-start)	Maintenance Schedule period = 'Pre-Start'	Create per equipment; items = checklist lines.
Maintenance Task (weekly/monthly)	Maintenance Schedule period = 'Weekly' / 'Monthly'	Set start/end dates and periodicity.
Completed maintenance	Maintenance Visit	Auto-created when a task is completed in the MES.

Assets: Assets → Asset → New. Create one asset per filling line / key equipment with the same ID used in MES Admin → Lines & Equipment (e.g. LINE-1-FILLER). Set maintenance team and depreciation only if finance requires it.

Part C — API Connection

C.1 Generating API Keys in ERPNext

The MES authenticates to ERPNext using per-user API tokens. Use a dedicated integration user (recommended: a user with System Manager or the roles listed in B.7) rather than a human user's keys.

Steps:
Log into ERPNext as a System Manager.

Go to Settings → User and open the integration user (e.g. 'mes-integration').

Scroll to the API Access section. If keys were never generated you will see a Generate Keys button — click it.

Copy the API Key and API Secret immediately — the secret is shown only once. Store them in your password manager.

Ensure the user is enabled and has the roles needed for every module the MES will touch (see B.7).

If the keys are later compromised, return to the same screen and click Generate Keys again to rotate them.

C.2 MES Connection Settings

Enter the ERPNext endpoint and credentials in the MES. These are stored server-side (never in the browser) once the backend is deployed.

MES navigation: Admin → Settings → Integrations → ERPNext

Setting	Example	Notes
ERPNext URL	https://erp.islandchill.example	Base URL only — no /api suffix.
API Key	1a2b3c4d5e6f7890	From C.1.
API Secret	••••••••••••	From C.1. Stored encrypted.
Company	Island Chill Beverages Ltd	Auto-filled after first successful connection.
Default Warehouse	ICB-Stores — Raw Material	Fallback for stock lookups.
Sync mode	Realtime / Scheduled / Manual	Realtime for production; scheduled for master data.
Pull interval (master data)	60 minutes	Users, warehouses, items, BOMs.
Retry attempts	5	Exponential backoff between attempts.
Dead-letter alert email	admin@islandchill.example	Notified when a sync job fails permanently.

C.3 Testing the Connection

In Admin → Settings → Integrations, click Test Connection. The MES calls GET /api/method/frappe.auth.get_logged_user and expects the integration user's email in response. A green banner confirms connectivity; a red banner shows the HTTP status and ERPNext error message.

Manual verification with curl:

```
curl -s -H "Authorization: token YOUR_KEY:YOUR_SECRET" \
https://erp.islandchill.example/api/method/frappe.auth.get_logged_user
```

Expected response:

```
{"message": "mes-integration@islandchill.example"}
```

Part D — Data Sync Workflows

This part describes exactly what happens in each sync flow, so administrators can verify behavior and diagnose mismatches. All flows assume the connection from Part C is healthy.

D.1 Users & Roles Sync

Direction: ERPNext → MES. Trigger: on user create/update in ERPNext (webhook) plus scheduled pull.

Step	Action	ERPNext Source
1	MES pulls users where custom field in_mes = 1	User list (GET /api/resource/User)
2	Role mapped via the B.7 table; MES permission set applied from the built-in role definition	User roles
3	New MES account created (active), or existing account updated; deactivated in ERPNext becomes inactive in MES	User enabled flag
4	MES Admin → Users shows synced users with 'ERPNext' source badge	—

Passwords never sync. Users sign into the MES with the same email; authentication can be delegated to ERPNext (OAuth) or handled by the MES login with local credentials — configurable in Admin → Settings.

D.2 Warehouse & Stock Sync

Direction: Bi-directional. Warehouses: ERPNext → MES. Stock movements: MES → ERPNext (submitted) and ERPNext → MES (balances).

Flow	What happens	ERPNext DocTypes
Warehouse master	Warehouses and types pulled hourly; MES zones update automatically	Warehouse, Warehouse Type
Opening stock	ERPNext Bin balances per warehouse-item pulled hourly; MES Warehouse page shows live qty, batch, expiry	Bin, Stock Ledger Entry
Goods receipt	MES Warehouse → Goods Receipt posts a Material Receipt stock entry	Stock Entry (Material Receipt)
Requisition picking	Requisition status 'picking' creates a DRAFT Material Transfer entry (reservation)	Stock Entry (draft)
Requisition delivered	Draft entry submitted; stock moves store → WIP; batches consumed FIFO with expiry check	Stock Entry (Material Transfer)
Production complete	WO 'Completed' posts a Manufacture entry: consume BOM items from WIP, produce FG into Finished Goods	Stock Entry (Manufacture)
QC rejection	Failed quantity moved to Quarantine or Scrap via Material Issue	Stock Entry (Material Issue)

Batch/expiry: items with 'Has Batch No' are consumed by FEFO (first-expired-first-out). The MES pick list shows lot numbers and expiry dates sourced from ERPNext batches; the Mixer must scan or confirm the lot before marking an item fetched.

D.3 BOM & Requisition Sync

Direction: BOMs ERPNext → MES; requisitions MES → ERPNext.

Step	Action
1	Planner approves a BOM in ERPNext (submit). MES pulls submitted BOMs on change.
2	MES Work Order creation requires a synced BOM; the BOM picker lists active ERPNext BOMs by item.
3	WO approved → MES auto-creates a BOM Requisition (draft) with a pick list generated from the BOM, scaled to batch quantity.
4	Mixer starts picking → draft Stock Entry created in ERPNext (reservation).
5	Mixer marks prepared → lot/expiry recorded; any substitution noted in the entry remarks.
6	Mixer delivers → Stock Entry submitted (Material Transfer, store → WIP). Requisition status becomes 'delivered'.
7	If WO is cancelled before delivery, the draft Stock Entry is cancelled and the reservation released.

D.4 Work Order & QC Sync

Direction: Bi-directional. Planners may create Work Orders in ERPNext (they sync to the MES), or in the MES (they sync to ERPNext).

MES Event	ERPNext Action
WO created in MES	Work Order created (status Draft). BOM and warehouse from the MES WO.
WO approved	Work Order submitted → Not Started.
Production started	Job Card created per operation; status In Process; start time stamped.
Stage completed / qty updated	Job Card time logs and completed qty updated.
WO completed	Job Cards closed; Manufacture Stock Entry posted; Work Order status = Completed.
QC check performed	Quality Inspection created with template, readings, result (Accepted/Rejected), inspector, timestamp.
Incident reported	Issue created with type/severity/priority; assigned per MES investigator.
Incident resolved	Issue status = Closed with resolution notes and root cause.
Maintenance task completed	Maintenance Visit created with completion details.

Because both systems can create Work Orders, the MES uses the ERPNext naming series with an 'MES-' prefix to avoid collisions (configurable in Admin → Settings → Integrations).

Part E — API Endpoint Reference

All endpoints are relative to the ERPNext base URL and use token authentication (Authorization: token API_KEY:API_SECRET). The MES sync engine uses exactly these calls.

Purpose	Method & Endpoint	DocType
Test connection	GET /api/method/frappe.auth.get_logged_user	—
List users	GET /api/resource/User?filters=[["in_mes","=",1]]	User
List warehouses	GET /api/resource/Warehouse?limit_page_length=500	Warehouse
Stock balances	GET /api/resource/Bin?filters=[["warehouse","=","WH-NAME"]]	Bin
List items	GET /api/resource/Item?limit_page_length=500	Item
List BOMs	GET /api/resource/BOM?filters=[["docstatus","=",1]]	BOM
Get single document	GET /api/resource/{Doctype}/{name}	any
Create document	POST /api/resource/{Doctype}	any
Update document	PUT /api/resource/{Doctype}/{name}	any
Cancel document	POST /api/resource/{Doctype}/{name} + {"docstatus": 2}	submittable
Submit document	POST /api/resource/{Doctype}/{name} + {"docstatus": 1}	submittable
Create stock entry	POST /api/resource/Stock Entry	Stock Entry
Create quality inspection	POST /api/resource/Quality Inspection	Quality Inspection
Create work order	POST /api/resource/Work Order	Work Order
Create issue	POST /api/resource/Issue	Issue
Server scripts / custom	POST /api/method/{dotted.path.to.method}	—

Example — create a Material Transfer Stock Entry:

```
POST /api/resource/Stock Entry { "stock_entry_type": "Material Transfer", "company": "Island Chill Beverages Ltd", "items": [{ "item_code": "RM-Water", "qty": 1650000, "uom": "ml", "s_warehouse": "ICB-Stores - Raw Material", "t_warehouse": "ICB-Production - WIP", "batch_no": "BW-2026-0412" }], "remarks": "MES Requisition req1 / WO-2025-001", "custom_mes_reference": "req1" }
```

The custom_mes_reference field is a custom field added to Stock Entry (and Quality Inspection, Issue, Work Order) via Settings → Customize Form. It stores the MES record ID for idempotency and traceability — the sync engine checks this field before creating to prevent duplicates.

Part F — Troubleshooting

Symptom	Likely cause	Fix
Test Connection fails with 401	Wrong API key/secret, or user disabled	Regenerate keys (C.1); confirm user is enabled in ERPNext.
Test Connection fails with 403	User lacks required roles	Assign roles per B.7; save the user record.
Connection timeout	Firewall / wrong URL / not HTTPS	Verify URL from the MES host: curl the health endpoint; open port 443.
Users not appearing in MES	in_mes flag not set, or pull job not running	Set custom field in_mes=1 on the User; check sync log in Admin → Settings.
Stock quantities differ between MES and ERPNext	Manual entries in ERPNext, or a failed push queued	Check the sync queue; run a manual Bin pull; reconcile via Stock Ledger report.
Requisition delivered but stock did not move	Draft Stock Entry failed validation (e.g. insufficient stock)	Open the draft entry in ERPNext; resolve the validation error; resubmit from the sync queue.
Duplicate stock entries after retry	custom_mes_reference missing or mismatched	Add the custom field to Stock Entry; confirm the sync engine reads it.
BOM not visible in MES	BOM is draft (docstatus 0)	Submit the BOM in ERPNext (only submitted BOMs sync).
QC inspection rejected in ERPNext	Reading outside template range or template not linked	Check Quality Inspection Template min/max; link template to the item (B.11).
Slow sync	Large Bin dataset pulled too often	Increase pull interval for master data; keep real-time only for stock movements.

Where to look in the MES:
• Admin → Settings → Integrations → Sync Log: every push/pull with status, record count, and error detail.
• Admin → Settings → Integrations → Dead-Letter Queue: transactions that failed after all retries; requeue or discard from here.
• Admin → Reports → Integration Health: success rate per module over time.
• Warehouse / BOM Requisition pages show a small ERPNext badge when a record has synced; a spinner indicates a pending push.

Document Summary

This guide covered the complete ERPNext integration for Island Chill MES: the architecture and module mapping (Part A), full step-by-step ERPNext configuration for administrators — company, warehouses, UOMs, items, users, BOMs, stock flow, workstations, quality, and maintenance (Part B), API key generation and MES connection settings (Part C), per-module data-sync workflows (Part D), the exact API endpoints used (Part E), and troubleshooting (Part F).

Keep ERPNext as the system of record for master data, let the MES drive shop-floor execution, and monitor the Sync Log daily during the first weeks of production use.